BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 8 — Magnetism

Total Marks: 65 | **Time Allowed: 2 hours 30 minutes** | **Version: 1**

General Instructions:

1. This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
2. **Section A** (12 marks): 12 MCQs. Mark the correct bubble. Time: 15 minutes.
3. **Section B** (33 marks): 11 short questions × 3 marks each. Answer *either* the main question *or* the OR alternative.
4. **Section C** (20 marks): 4 long questions × 5 marks each. Answer *either* the main question *or* the OR alternative.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	The basis of magnetism, according to domain theory, is the motion of:	Protons	Neutrons	Electrons	Nuclei	ⒶⒷⒸⒹ
2	A magnetic domain contains roughly how many atoms?	10^3	10^6	10^{12}	10^{20}	ⒶⒷⒸⒹ
3	Magnetic field lines of a bar magnet originate from the:	South pole	Centre	North pole	Both poles equally	ⒶⒷⒸⒹ
4	The SI unit of magnetic field strength B is:	Weber	Tesla	Ampere	Henry	ⒶⒷⒸⒹ
5	A material that becomes a magnet when placed inside a magnetic field is called a/an:	Permanent magnet	Induced magnet	Diamagnetic material	Hard magnet	ⒶⒷⒸⒹ
6	Which is the most commonly used method for magnetizing metals?	Stroking method	Hammering method	Heating method	Solenoid method	ⒶⒷⒸⒹ
7	Which material is best suited for making an electromagnet?	Steel	Cobalt	Soft iron	Nickel	ⒶⒷⒸⒹ
8	Diamagnetic materials have a net magnetic field per atom that is:	Very large	Small but non-zero	Zero	Equal to 1 T	ⒶⒷⒸⒹ
9	Earth's magnetic and geographical poles are inclined at an angle of:	0°	11.3°	23.5°	90°	ⒶⒷⒸⒹ
10	Which of the following is a ferromagnetic material?	Silver	Copper	Aluminium	Nickel	ⒶⒷⒸⒹ
11	If a bar magnet is cut in half, it will become:	A monopole	Unmagnetized	Two complete magnets	A weaker magnet	ⒶⒷⒸⒹ
12	Earth's magnetic field is caused by:	Rotation of Earth	Spinning of Earth	Pull of the Sun	Motion of ions in the core	ⒶⒷⒸⒹ

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

Q.2: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Part	Question	Marks	OR	OR Question	Marks
(i)	Define the **domain theory of magnetism**. What is a domain? How do domain arrangements differ in magnetized vs unmagnetized material?	**3**	**OR**	Describe how **electrons in an atom** produce magnetism. Why is the contribution of the spinning nucleus neglected?	**3**
(ii)	Define **magnetic field**. State its SI unit. Explain how the direction of the magnetic field at any point is determined.	**3**	**OR**	What are **magnetic field lines**? State any three properties of the field lines of a bar magnet.	**3**
	Explain **relative strength of a magnetic**			What is **magnetic shielding**? Explain the	

(iii)	**field** as shown by field lines. What does close spacing of field lines indicate?	**3**	**OR**	neutral zone. Name three materials used as magnetic shields and why their corners are rounded.	**3**
(iv)	Define **induced magnetism**. Describe the **stroking method** step by step. Which pole forms at the end where stroking finishes?	**3**	**OR**	Describe the **hammering method**. For which material is it mainly used? How can its magnetization effect be increased before hammering?	**3**
(v)	Explain how the **heating method** demagnetizes a material. What is the "magnetic Seebeck effect" mentioned in the chapter?	**3**	**OR**	Describe the **solenoid method** of magnetization. Why does DC magnetize in one direction while AC demagnetizes the material?	**3**
(vi)	Write the formula $\mathbf{B = \mu_o nI}$. Define each symbol with its unit and state the value of μ_o for vacuum.	**3**	**OR**	A solenoid has 200 turns, length 50 cm, current 2 A. ($\mu_o = 4\pi\times10^{-7}$ N A^{-2}) Calculate the magnetic field **B** inside.	**3**
(vii)	Differentiate between **temporary magnets** and **permanent magnets**. Give two examples of each.	**3**	**OR**	Why is **soft iron** used for electromagnets while **steel** is used for permanent magnets? Relate to soft and hard magnetic materials.	**3**
(viii)	Differentiate between **diamagnetic** and **paramagnetic** materials (net field per atom, behaviour in external field, two examples each).	**3**	**OR**	Define **ferromagnetic materials**. State three properties distinguishing them from paramagnetic materials. Give two examples.	**3**
(ix)	Differentiate between **magnetic** and **non-magnetic materials**. Give three differences and one example of each.	**3**	**OR**	Can two magnetic field lines ever intersect? Justify your answer with reference to what the direction of the field at that point would mean.	**3**
(x)	Describe **Earth's magnetic field**. What is the dynamo effect and how does it generate Earth's field?	**3**	**OR**	Why are Earth's **geographical and magnetic poles not the same**? By what angle are they offset and what is the practical effect on a compass?	**3**
(xi)	Explain how **speakers** use a permanent magnet and an electromagnet to produce sound. What material is used as the permanent magnet?	**3**	**OR**	Explain the principle of **magnetic recording**. Name three recording media. State two types of devices that use magnetic recording.	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Explain the **domain theory of magnetism**: (a) how electrons produce magnetism at the atomic level, (b) definition and typical size of a domain, (c) domain arrangement before and after magnetization, (d) what happens to domains when a magnetized material is heavily heated.	**5**	**OR**	Define **magnetic field** and **field lines**. Explain: (a) plotting with compass and iron filings, (b) direction of lines inside and outside a bar magnet, (c) what spacing of lines indicates, (d) what a neutral zone is and how it is formed by two N-poles placed side by side.	**5**
3(ii)	Explain **three methods of magnetization** (stroking, hammering, solenoid). For the solenoid write $B = \mu_o nI$ and define all terms. Calculate B for a solenoid: N = 100 turns, L = 25 cm, I = 3 A, $\mu_o = 4\pi\times10^{-7}$ N A^{-2}. State which method is most common and why.	**5**	**OR**	Compare **permanent magnets** and **electromagnets** under four headings: material used, how magnetism is produced, whether strength can be varied, and whether poles can be changed. List four applications of permanent magnets and four of electromagnets.	**5**
3(iii)	Classify magnetic materials as **diamagnetic, paramagnetic, and ferromagnetic**. For each: (a) net field per atom, (b) behaviour in an external field, (c) whether magnetism is retained after field removal, (d) two examples. State which type the chapter calls "non-magnetic material."	**5**	**OR**	Using five textbook points, differentiate **magnetic from non-magnetic materials**. Then explain **magnetic shielding**: define it, state why it is needed, explain the neutral zone, name two shield materials, and explain rounded corners.	**5**
3(iv)	Describe **Earth's magnetic field**: (a) dynamo effect and source, (b) resemblance to bar magnet, (c) angle between geographical and magnetic poles, (d) protection from cosmic rays, (e) how Aurora forms.	**5**	**OR**	Explain two magnet applications: (a) **magnetic recording** (process step by step, three recording media, two device types), and (b) **speakers and microphones** (role of each magnet; why microphone is the reverse of a speaker).	**5**

Invigilator's Signature: ________________ Candidate's Signature: ________________

Chapters Covered: Unit 8 — Magnetism **-- End of Question Paper --**